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Mental Fatigue Negatively Impacts Cognitive and Physical Performance Outcomes: A Large-Scale Randomized Crossover Trial

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ABSTRACT

HABAY, J., Y. L. ARENALES ARAUZ, M. PROOST, E. SCHAMPHELEER, E. LATHOUWERS, K. DE PAUW, N. PATTYN, J. VAN CUTSEM, and B. ROELANDS. Mental Fatigue Negatively Impacts Cognitive and Physical Performance Outcomes:ALarge-Scale Randomized Crossover Trial. *Med. Sci. Sports Exerc.*, Vol. 58, No. 2, pp. 225–241, 2026. **Purpose:** Although many studies have demonstrated that prolonged cognitive load can impair subsequent cognitive and physical performance, commonly described as mental fatigue, more recent findings suggest that the relationship is more nuanced. Moreover, significant heterogeneity between different smaller studies makes it difficult to summarize definitive conclusions. The aim of the present study was to provide a more robust examination of the detrimental effects of sustained cognitive demands on cognitive and physical performance fatigability using a large sample within a controlled environment. **Methods:** One hundred seventeen participants (57 female; 32 ± 9 yr) were included in this randomized counterbalanced crossover experiment (NCT05576935) consisting of familiarization, intervention (45-min individualized Stroop task), and control (45-min self-chosen documentary) sessions. Performance was evaluated using a 6-min GoNoGo task and a 20-min cycling time trial. Multiple secondary outcomes, such as rate of perceived exertion (RPE), feelings of fatigue, and motivation, were incorporated to explore their relationship to possible behavioral effects. **Results:** There was a significant worsening of GoNoGo reaction time between conditions ($P < 0.001$; $\eta_p^2 = 0.07$). A trend toward significance was found regarding the negative influence of the Stroop task on time trial distance ($P = 0.059$; $d = 0.20$), which was linked to a significant decrease in cadence ($P = 0.042$; $d = 0.22$) in the intervention condition. Moreover, the feelings of physical fatigue ($P < 0.001$; $\eta_p^2 = 0.01$), mental fatigue ($P < 0.001$; $r = 0.37$–0.47), and RPE ($P = 0.002$; $\eta_p^2 = 0.01$) remained higher during the time trial in the intervention condition compared with the control condition. **Conclusions:** In the present study, prolonged cognitive effort impaired specific cognitive and physical performance outcomes, consistent with the characteristics of mental fatigue. Secondary outcomes show an important role for the feeling of fatigue in the determination of physical performance. **Key Words:** FATIGUE, STROOP TASK, RESPONSE INHIBITION, ENDURANCE PERFORMANCE, GONOGO, TIME TRIAL, RPE

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Fatigue is a complex and daily occurring phenomenon, with a prevalence exceeding 20% in the general healthy adult population (1,2). It is nondiscriminating, impacting all ages, genders, and ethnicities, although with varied intensity (2). Because of the multidimensional nature of this phenomenon, multiple definitions and interpretations of fatigue are present in the literature (3,4). Fatigue has been described as an experimental concept, a symptom, a risk, a cause, and a consequence (4). To aid in its interpretation, the concept of fatigue has been categorized into various constructs, based on criteria such as induction methods (e.g., physical and mental) or origin (e.g., central and peripheral) (4). Although the use of these distinctions is somewhat controversial in the spirit of embracing human complexity (3,5), focusing on different aspects of the same phenomenon could actually aid us in better understanding it, especially in relation to states as intricate as fatigue.

Moreover, the use of fatigue components is well accepted in practice (6,7) and has been integrated into several theoretical frameworks (4,8,9). The present study focuses specifically on fatigue that is induced by prolonged cognitive exposure, commonly referred to as mental fatigue. As it is not the aim of the study to provide a final answer on the taxonomy of fatigue, it is imperative that multiple views of the argument are presented fairly throughout the present study. Therefore, although the fundamentals of the article are based on the model by Enoka and Duchateau (3), the theories behind the phenomenon defined as mental fatigue are integrated throughout the manuscript whenever relevant.

Mental fatigue is defined in the literature as a psychobiological state that arises during prolonged demanding cognitive activity and results in an acute feeling of tiredness and/or a decreased performance capacity (10,11). Research interest has surged in the past 15 yr, establishing mental fatigue research as its own distinct field within fatigue literature (12,13). This is related to the finding that prolonged cognitive activity negatively impacts both cognitive (14) and physical performance (15). This negative effect on physical performance has been supported by a variety of systematic reviews, with evidence of decrements in multiple performance types, such as (strength) endurance, motor skills, and balance performance (10,11,16–20). Interestingly, the rate of perceived exertion (RPE) has been proposed as the main driver behind these performance changes, as traditional peripheral measurements such as heart rate (HR) remain unchanged under mentally fatiguing conditions (11). However, recent insights have revealed contradictory results regarding the cognitive load effects on human performance. More psychologically oriented research found limited evidence of a negative effect on cognitive performance (21–23). Regarding physical performance, partial replication studies, additional analyses of existing studies, and new investigations seem to indicate a nonsignificant effect compared with the effect sizes that are represented in the literature (24–27). Although criticisms of these recent investigations are also warranted (28), they nonetheless highlight important issues for the mental fatigue research field. This, in turn, limits real-life application and integration of solutions to deal with this type of fatigue.

The primary challenge in drawing firm conclusions about the general effects of mental fatigue induced by prolonged cognitive exposure on human performance lies in the significant heterogeneity across studies investigating this topic (10,16). This is to be expected, as methodologies have advanced considerably since the first study performed by Marcora et al. (15). Moreover, different authors and research groups will also look at this distinction of fatigue from different perspectives, enriching our understanding but complicating the integration of findings. Substantial interindividual variability in responses to mental fatigue (10) further obscures the conclusions drawn in systematic reviews and meta-analyses (10,11,19,24). It should also be noted that the impact of mental fatigue is notoriously difficult to measure, and it is still not possible to clearly "confirm" a state of mental fatigue. To avoid presenting mental fatigue as an established fact and to minimize bias, the term is used in this paper solely when referring to the specific phenomenon as described in previous literature.

A rigorously controlled study based on state-of-the-art research guidelines and methodologies including a large population sample will be able to provide more robust evidence of the influence of prolonged cognitive exposure on human performance. The present large-scale study was also conducted within a controlled environment as a solution to the heterogeneity present in previous evidence. The primary aim was twofold: first, to assess the effect of prolonged intense cognitive load on cognitive performance fatigability (i.e., a decline in an objective measure of cognitive performance over a discrete period (3), measured using a GoNoGo task; and second, to assess the influence of the same cognitive load on physical performance fatigability (i.e., a decline in an objective measure of physical performance over a discrete period (3), represented by a single measure of whole-body dynamic endurance performance. Secondary outcomes include a variety of subjective, behavioral, and physiological measures, which will be used to provide context for any observed effects on performance. We hypothesized that prolonged individualized cognitive exposure would have a significant negative effect on physical (11,19) and cognitive (14) performance based on previous literature.

MATERIALS AND METHODS

The present study was designed in accordance with the Consolidated Standards of Reporting Trials (CONSORT) 2010 extension for randomized crossover trials (29). The report adheres to the CHecklist for statistical Assessment of Medical Papers (the CHAMP statement) (30). The present study protocol and its procedures were approved by the Medical Ethics Committee of the UZ Brussel (B.U.N. 1432022000084) in accordance with the Declaration of Helsinki. Informed consent was obtained from the participants. The trial was registered on ClinicalTrials.gov (NCT05576935). The present article is part of a larger research project determining the individual response to fatigue (Grant number FWO: 11J6323N).

Participants

A sample size calculation using the G*power software (version 3.1.9.7.) determined that 101 participants needed to be included in the present investigation. This number was calculated using the matched-pairs *t* test

analysis, with the effect size suggested in the meta-analysis of Brown et al. (19) ($d = 0.384$; effect of prior cognitive exertion on "aerobic performance"), a power value of 0.95, an α of 0.05, and taking into account a possible dropout of 10%.

Participants were required to register through a Microsoft Forms online link. After this registration, participants were sent a form checking their mental health status, consisting of the multidimensional fatigue inventory (31), the Burnout Assessment Tool (32), and the Beck Depression Index-II (33) (see Supplemental Digital Content 1, http://links.lww.com/MSS/D299). Participants were unaware of the goal of this study and were told that we examined cerebral activity differences during discrepant cognitive and physical trials.

Eligible participants were healthy adults between 18 and 50 yr old. Participants were excluded if they used drugs, were pregnant, and suffered from color vision deficiencies or any acute or chronic health conditions (both physical and mental). During the trial, the inclusion criteria of participants were refined to ensure balanced groups based on age and sex. Before each experimental trial, participants were required to abstain from exhaustive physical activity the day before, and any type of caffeine the evening before and the day of the trial. They were asked to eat a similar meal the evening before and the morning of the trial, and to get at least 7 h of sleep.

Trial Design and Measurement Procedures

A randomized, single-blind, controlled, counterbalanced crossover design was applied. The experimental trials (i.e., intervention and control) were separated by at least 3 d to ensure full recovery (34) and were conducted around the same time of day (between 7:00 and 13:00). All experimental trial procedures took place at the same location (MFYS/BLITS VUB, Brussels, Belgium) in a soundproof climate chamber (Weiss Technik Belgium, Liedekerke, Belgium) providing a controlled atmosphere (20°C, 40% humidity).

When arriving for the intake/familiarization trial, participants received a brief explanation and signed the informed consent form. Afterward, participants visited the medical doctor for a general clinical examination, a blood draw, and a rest electrocardiogram. Thereafter, participants performed a maximal incremental cycling exercise test (80 W + steps of 30 W) to assess physical performance level (35,36). After this, a personal screening of participants was performed. A detailed description of the used questionnaires and tasks can be found in Supplemental Digital Content 1, http://links.lww.com/MSS/D299. Finally, the participants were able to familiarize themselves with all questionnaires and tests used during the study protocol. These questionnaires and tests included the following: the mental fatigue visual analogue scale (M-VAS), the verbal subjective feeling of mental fatigue, boredom-VAS (B-VAS), motivation-VAS (Moti-VAS), National Aeronautics and Space Administration Task Load Index 4 (NASA-TLX), Brunel Mood Scale (BRUMS), Karolinska Sleepiness Scale (KSS), RPE (RPE-100), the verbal subjective feeling of physical fatigue, the cognitive intervention task (i.e., Stroop max task), and the different performance tasks (i.e., GoNoGo task and cycling time trial).

The next visits were the intervention and control trial. After checking adherence using the pretest checklist, participants were fitted with an HR monitor and an electroencephalography (EEG; BrainAmp, Acticap slim/snap, Brain Products, Munich, Germany) device. An EEG baseline with eyes open (2 min) and eyes closed (2 min) was recorded. This was followed by a bundle of questionnaires (i.e., M-VAS and Moti-VAS in that order) and by the "pre" cognitive performance task (i.e., GoNoGo). Afterward, different questionnaires were again administered (i.e., M-VAS, NASA-TLX, BRUMS, KSS, and the Moti-VAS) before starting the intervention/control task (i.e., a 45-min modified Stroop task or emotionally neutral documentary). After this task, a selection of questionnaires was repeated (i.e., B-VAS, BRUMS, KSS, and the Moti-VAS). The GoNoGo trial was again performed before another bundle of questionnaires (i.e., M-VAS, NASA-TLX, and Moti-VAS). The measurement ended with a 20-min cycling time trial. Figure 1 shows a schematic overview of the overall study protocol. Considering the vast amount of data, we chose not to include EEG outcomes in the present analyses.

Stroop max task. The Stroop max task was designed to ensure an individualized cognitive load for every participant (37–40). As with all cognitive tasks, the Eprime software (Psychology Software Tools, Pittsburgh, PA) was used to construct the task. Participants were required to perform multiple Stroop tasks (for procedure, see the Effects on physical performance fatigability section) during the familiarization, each having a duration of about 5 min (72 samples, 2 cycles of 36 samples), with a progressively decreasing stimulus presentation time and identical varying interstimulus time (1100–1900 ms). The first Stroop max task was a warm-up with a stimulus presentation time of 1500 ms. Afterward, participants started with the 1100-ms task. The task increased in difficulty if they reached an accuracy of at least 85% or more (41). Presentation time decreased in 100-ms intervals (1100, 1000, 900 ms …), with 600 ms being the lowest presentation time that could be reached. Participants were allowed three trials for every level, and five second tries in total. If participants failed, the last successfully reached level was chosen as the stimulus presentation time in the experimental Stroop task. In the case of failing at 1100 ms, a stimulus presentation time of 1200 ms was provided.

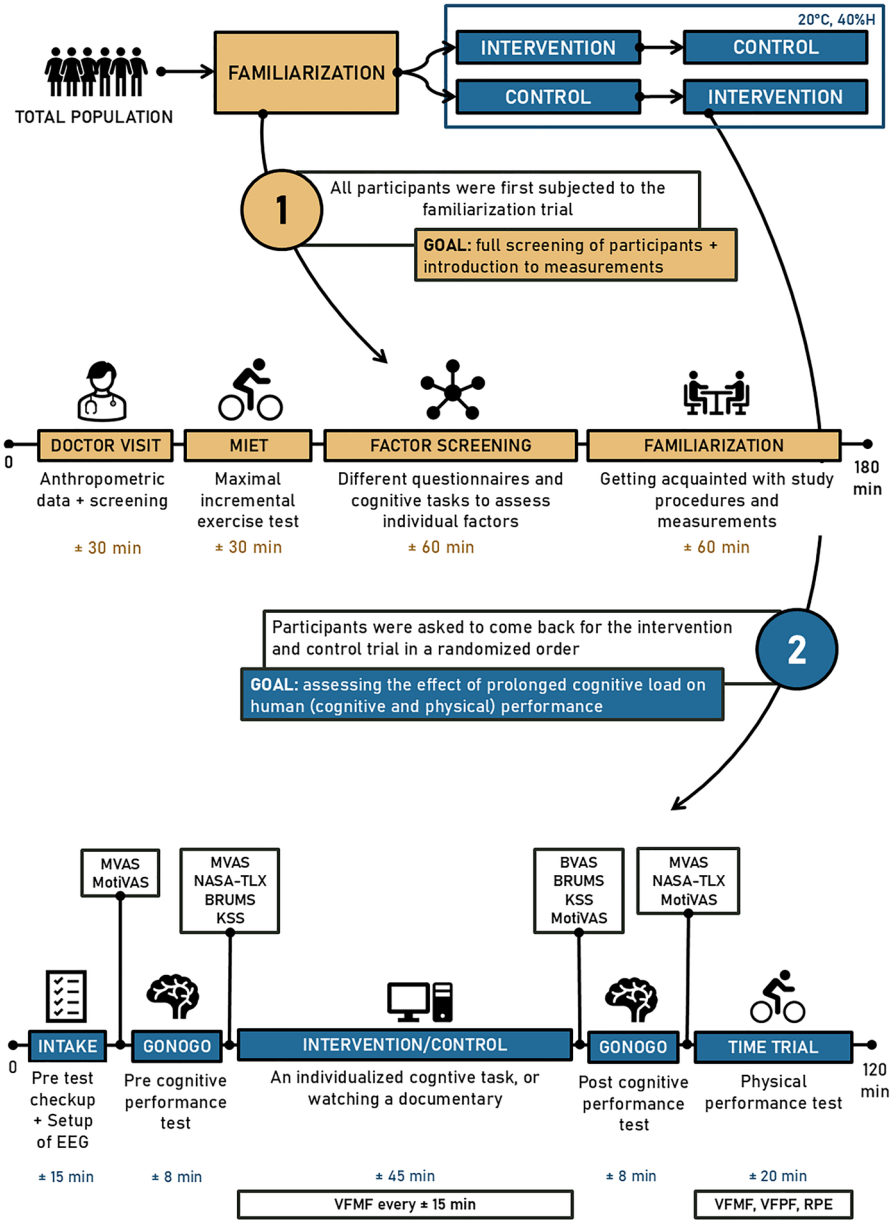

FIGURE 1—Overview of the complete study protocol, with a detailed description at the familiarization and experimental/control trial (%H = percentage of humidity; BVAS = boredom visual analogue scale; MIET = maximal incremental exercise test; MotiVAS = motivation visual analogue scale; MVAS = mental fatigue visual analogue scale; NASA-TLX = National Aeronautics and Space Administration Task Load Index 4, VFMF = verbal feeling of mental fatigue; VFPF = verbal feeling of physical fatigue).

Intervention task. A modified Stroop task (37,42,43) of approximately 45 min, partitioned in three blocks of 360 stimuli, was used as the intervention task. In this task, four color names ("rood," "blauw," "groen," and "geel") were presented in colored letters on a computer screen, one word at a time. The participant was required to indicate the color of the word, ignoring the meaning of the word. However, if the color of the word was red, participants needed to react to the meaning of the word. The word presented and its color were randomly selected by the Eprime software (100% incongruent), with all incongruent word-color combinations being equally common. Each word was presented on a screen with a black background in 34-point font. The interstimulus interval varied between 1100 and 1900 ms, with a mean of 1500 ms. The stimulus presentation time was based on the maximal performance on the Stroop max task (see the Effects on cognitive performance fatigability section). Subjects were instructed to respond as quickly and accurately as possible.

Control task. In the control condition, participants watched an emotionally neutral documentary of the same duration as the intervention task. A list of different documentaries that were available on the streaming site

Disney+ was provided for participants to pick the film of their choice. This list of documentaries can be found in Supplemental Digital Content 1, http://links.lww.com/MSS/D299.

Time trial. The influence of the intervention on physical performance fatigability was assessed using a 20-min cycling time trial performed after the intervention/control task. The trial took place on the Cyclus 2 cycle ergometer (version 4.2, Leipzig, Germany) with a common race bike (Trek Domane AL 2 Rim) attached. Before the task, blood lactate, blood glucose, and verbal feeling of physical fatigue (from 0 to 100) were assessed. Then participants started with a 3-min warm-up at a fixed intensity of 80 W, where they got a brief reminder of the performance task. The time trial itself was 15 min, where participants were allowed to freely choose the resistance throughout the trial. They were instructed to cover as much distance as possible. The only information that was available for the participants was the time that was left during the trial. Every 3 min, HR was assessed, and the feeling of physical fatigue, RPE, and feeling of mental fatigue were evaluated verbally. The first time these questions were asked was during the final minute of the warm-up, and participants were instructed they could already change their resistance to have a rolling start. The last time the questions were asked was 1 min before the end of the trial, so that participants could focus on a possible end sprint. When the trial started, a fan (Wahoo KICKR© Headwind Ventilator) ensured adequate cooling of the participant during the physically exhaustive task. After this, a 2-min fixed intensity cool-down of 60 W commenced, where lactate and blood glucose were sampled again, and the three questions were asked a final time after 1 min. Participants did not receive any motivational feedback at any point during the task.

GoNoGo task. The effects of Stroop task exposure on cognitive performance fatigability were quantified using a modified GoNoGo task (14,44,45) before and after the intervention/control trial. During the task, two objects were presented on a computer screen: a geometric figure (square or triangle) and an arrow (pointing either left or right). The figures (area of $\pm 42\,cm^2$) and arrows (length of 6.5 cm) were presented in white on a black background on a 17.3-inch laptop screen. When a square was presented (i.e., "Go" stimulus), participants were asked to react as fast and accurately as possible to the arrow with the appropriate keys on the keyboard, so either "right" or "left." When a triangle was presented (i.e., "NoGo" stimulus), participants were instructed to withhold any response and to wait until a new stimulus appeared. This meant that the following outcomes could be collected: reaction time (RT) on the Go stimuli and accuracy (ACC) on both the Go and NoGo stimuli. The Go to NoGo ratio was set at 80/20, with three cycles of 60 stimuli (with 48 "Go" and 12 "NoGo" stimuli within each cycle). These stimuli were randomly selected within one cycle and presented on screen for 500 ms with varying interstimulus intervals ranging between 1100 and 1700 ms (total task duration of about 6 min).

Outcomes of the Present Trial

Primary outcomes: indicators of physical and cognitive performance. The cognitive task resulted in three specific primary outcomes: Go RT (ms), and Go and NoGo ACC (presented in a ratio from 0 to 1). The primary outcome of the physical performance task was total distance covered (in km).

Secondary outcomes: manipulation checks, additional indicators of performance, and subjective outcome measures. The subjective presence of mental fatigue was evaluated using the M-VAS. Meanwhile, behavioral effects of the intervention task were assessed using the outcomes of the Stroop task. The M-VAS was used before and after the first and second GoNoGo task. Participants were required to answer the question "how mentally fatigued do you feel?" on a scale of 10 cm where one end represented "not at all" and the other "completely exhausted" (46). Stroop task ACC (presented in a ratio from 0 to 1) and RT (in ms) were collected, averaged for every block, and divided for normal incongruent (i.e., green, yellow, and blue) and inhibitory incongruent (i.e., red) stimuli. To examine the Stroop effect (47), a difference score for RT and ACC was calculated between the normal and inhibitory stimuli (formula: meanNormal − meanInhibitory). The following outcomes were also collected before and after the Stroop task: mood (measured by the BRUMS [48]; results divided in six subscales each ranging from 0 to 16 arbitrary units (AU)), sleepiness (measured by the KSS [49]; a scale ranging from 1 = "extremely alert" to 10 = "extremely sleepy, falls asleep all the time"), and boredom (measured by the B-VAS [50] only after the Stroop task; 0–100 in AU).

Secondary outcomes collected during the physical performance task included total workload (in kJ), average and maximal power output (in watts), and mean cadence (in rpm). Moreover, the pacing of the power output (in watts) and cadence (in rpm) was also utilized as an outcome. A text file was gathered, which was recalculated to provide a mean for every second of the trial (i.e., 1080 seconds). Subjective outcomes during the time trial included the feeling of mental fatigue ("How mentally fatigued do you feel on a scale of 0 to 100?"; 0–100 in AU), the feeling of physical fatigue ("How physically fatigued do you feel on a scale of 0 to 100?"; 0–100 in AU), and RPE ("How heavy do you perceive the task to be on a scale of 0 to 100?"; measured using the RPE 100 scale of Borg et al. [51]). Before the trial, the motivation was measured using a Moti-VAS (52) ("How motivated

do you feel to perform the next task?"; 0–100 in AU). Physiological outcomes included HR (Polar H9; in beats per minute), blood lactate (BIOSEN 5030, Magdeburg, Germany; in $mmol \cdot L^{-1}$), and blood glucose (Bayer Contour, Leverkusen, Germany; in $mg \cdot dL^{-1}$).

Meanwhile, mental workload (measured by the NASA-TLX 4 [53] after the GoNoGo; results divided in four subscales each ranging from 0 to 100 AU in intervals of 5) and motivation (measured using the Moti-VAS before the GoNoGo; 0–100 in AU) were collected to better understand possible cognitive performance differences between intervention and control. The values of the NASA-TLX were summed and divided by the number of scales (= 4) to achieve a raw score (53,54).

Data Processing and Statistical Analyses

All statistical tests were performed using R (version 4.3.1.) through the use of R studio (version 2024.04.2). The code can be found in Supplemental Digital Content 2, http://links.lww.com/MSS/D300. All raw data can be found in Supplemental Digital Content 3, http://links.lww.com/MSS/D301, and Supplemental Digital Content 4, http://links.lww.com/MSS/D302. All data are presented using mean ± standard deviation, unless stated otherwise because of skewness. The significance value was set at 0.05 for all analyses. Results were defined as trends to significance when P values were situated between 0.05 and 0.1. Different ranges were adopted to indicate the magnitude of the following effect sizes: Cohen's d (<0.2 = trivial; 0.2–0.6 = small; 0.6–1.2 = moderate; 1.2–2.0 = large; >2.0 = very large), and partial eta square (~0.01 = small; ~0.06 = medium; ≥0.1 = large). When specific effect sizes could not be provided, odds ratios or original estimates were given to compare analyses within the same population. All linear models were performed with the same fixed (task = Stroop task vs. documentary; and time = pre vs. post/comparison over multiple time points) and random (participant number) effects, unless stated otherwise. When interaction effects were found in the linear models, *post hoc* tests were performed. All pairwise comparisons confidence intervals (CIs) and estimates can be found in Supplemental Digital Content 1, http://links.lww.com/MSS/D299.

If the data of one specific variable was entirely missing for one or multiple trials, it was excluded from further analysis. Only single missing data points within one visit were added using linear interpolation. The total data yield for all variables is detailed in Supplemental Digital Content 1, http://links.lww.com/MSS/D299. Outliers were identified using the Z score method, with values exceeding absolute Z score above three flagged as outliers. Each outlier was subsequently evaluated to determine whether it resulted from an imputation error or another identifiable cause. Imputation errors were fixed, and only significant outliers with a clear exclusion reason (e.g., a very high or very low subjective value, even when compared with other time points within the same session and other sessions) were excluded from further analysis. Assumptions of normality were checked visually using Q-Q plots and histograms, and statistically using the Shapiro–Wilk test. When assumptions of normality were not met, attempts were made to transform the data using established methods (i.e., square root (sqrt), log, and inverse in that order). The following variables were transformed: time trial mean power (sqrt), time trial maximal power (log), motivation before the time trial and GoNoGo tasks (sqrt), the feeling of physical fatigue during the time trial (sqrt), the RPE (sqrt), and blood glucose (log). When data were not normally distributed and could not be transformed, nonparametric equivalents were utilized. Assumptions of homoscedasticity were examined when necessary through visual inspection of residual plots. In case of potential heteroscedasticity, we applied cluster-robust (CR2) standard errors for generalized linear models to obtain unbiased estimates and valid inference. The proportional odds assumption of the ordinal logistic regression models was evaluated using the Brant test.

The distance of the time trial, as well as all other mean outcomes, were analyzed using a paired samples t test (fixed effect: task). Linear mixed models were performed for the pacing, subjective, and physiological values taken during the time trial. The subjective values of mental fatigue collected during the time trial needed to be analyzed using nonparametric equivalents (time: Friedman, task: multiple Wilcoxon signed-rank tests). For pacing values, the warm-up and the cool-down were not included in the analyses as these phases were performed at a fixed intensity. Similarly, the cool-down was also not included in the analyses surrounding the subjective data collected during the time trial.

A mixed linear model was used to evaluate the influence of Stroop task exposure on Go RT, and motivation and mental workload linked to the GoNoGo task. The ACC of the Go and NoGo outcomes was assessed using a β regression (fixed effects: task and time). As the second research aim (i.e., the influence of the Stroop task on cognitive performance fatigability) contains three specific outcomes, corrections for multiple testing were applied using the Bonferroni method (significant if $P < 0.05/3$). When interaction effects were found in the linear models, *post hoc* paired samples t tests were performed.

As the data were not normally distributed, the results of the M-VAS were divided into three specific parts: baseline (before the first GoNoGo trial), intervention/control (before and after the intervention/control trial), and proof of induction (before the time trial). Wilcoxon signed-rank tests were performed to assess effects of both time and task. Stroop RT was evaluated using a linear mixed model. Stroop ACC values were needed to be

analyzed using separate Wilcoxon signed-rank (differences in types between blocks) and Friedman (differences in blocks within one type) tests. The Stroop effect was analyzed for both outcomes using a linear mixed model (fixed effects: time; random effect: participant number). The tension, anger, depression, and confusion subscales of the BRUMS were evaluated using β regressions (fixed effects: task and time). As the fatigue and vigor subscales were not normally distributed, separate Wilcoxon signed-rank tests were utilized to determine separate effects of task and time. The KSS values were modeled using ordinal regression (fixed effects: task and time), and the B-VAS values were compared using a Wilcoxon test.

RESULTS

Flow of Participants and Participants' Characteristics

Eventually, 281 participants filled in the registration questionnaire, of which 104 participants completed all experimental steps (Fig. 2).

Table 1 shows the different means and standard deviations of all variables that were collected during the familiarization trial.

Four participants were excluded from the overall data analysis because of unusually high overall Pittsburgh Sleep Quality Index (PSQI) values (ranging from 9 to 14) (55).

Effect of Prolonged Intense Cognitive Exposure on Performance and Related Outcome Measures

Table 2 displays the descriptive values for all outcomes collected during the intervention and control trial, as well as the results from the general statistical models.

Effects on cognitive performance fatigability. A schematic overview of the effect of Stroop task exposure on cognitive performance is depicted in Figure 3.

There was a significant interaction effect between task and time for the RT on the Go stimuli ($P<0.001$). *Post hoc* paired samples t tests showed no difference in values between task conditions before the intervention/control task ($t=-0.46$; $P=0.645$; $d=0.05$) and when

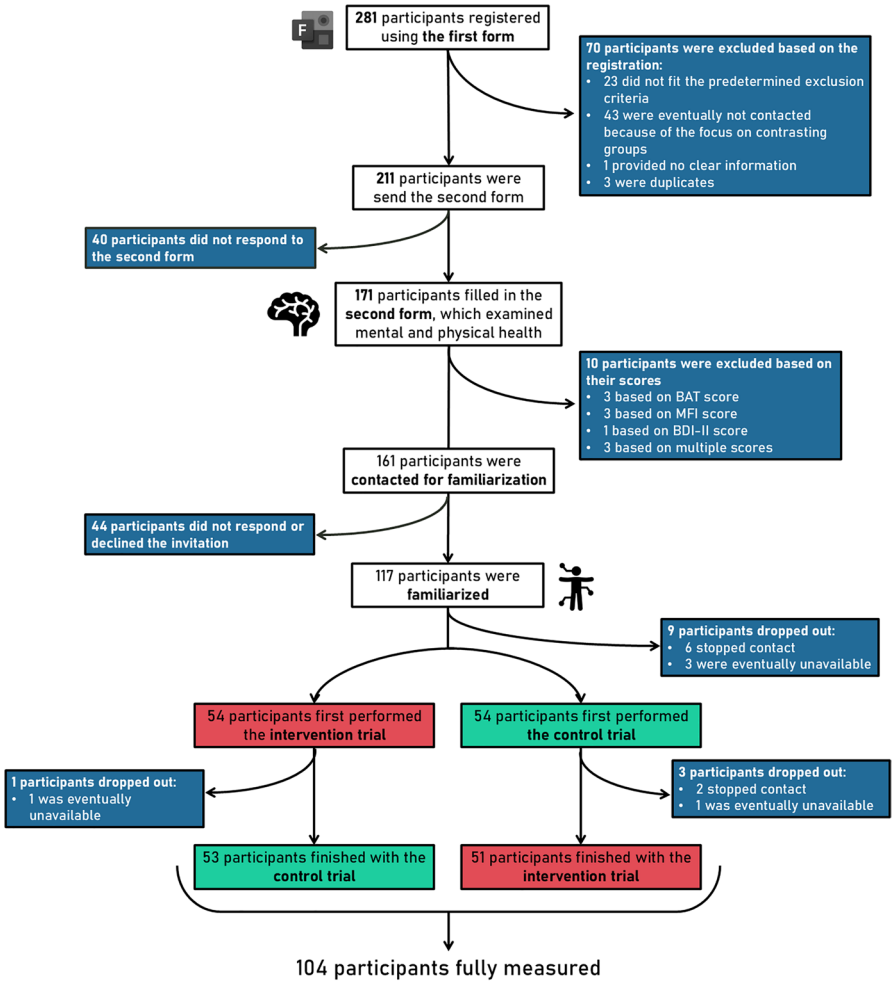

FIGURE 2—Consort figure portraying the participant flow throughout the trial (BAT = Burnout Assessment Tool; BDI-II = Beck Depression Inventory; MFI = Multidimensional Fatigue Index).

TABLE 1. Overview of means and standard deviations of collected baseline individual features.

Numerical Data		
Variable	***n***	**Mean ± SD**
Age (yr)	117	32 ± 9
Length (cm)	117	175 ± 9
Weight (kg)	117	69.3 ± 11.7
BMI (kg·m^{-2})	117	22.64 ± 2.63
Fat percentage (%)	116	20.6 ± 7.5
Relative $\dot{V}O_{2max}$ (mL·kg^{-1}·min^{-1})	116	48 ± 9
Absolute $\dot{V}O_{2max}$ (L·min^{-1})	116	3.32 ± 0.87
Peak power output (W)	117	250 ± 64
IPAQ (MET-min·wk^{-1})	116	3895 ± 2668
IPAQ (kcal·wk^{-1})	116	4593.46 ± 3308.53
Response Inhibition (SART Acc.)	117	0.57 ± 0.20
Working memory (2BACK Acc.)	117	0.83 ± 0.10
Attention (PVT RT)	117	289.81 ± 29.32
Trait anxiety (STAI)	116	32 ± 7
Trait Self Control (BSS)	116	4 ± 1
Mental Toughness (MT)	117	5 ± 1
Sleep Quality (PSQI)	117	4 ± 2
Caffeine consumption (mg·wk^{-1})	117	1470.74 ± 1372.51

Categorical data		
Variable	**Group**	***n***
Sex	Men	60
	Women	57
Performance levels based on $\dot{V}O_{2max}$ (De Pauw and Decroix)	Type 1	29
	Type 2	54
	Type 3	24
	Type 4	8
	Type 5	2
Performance levels based on level within practiced sport (McKay)	Tier 1	27
	Tier 2	47
	Tier 3	31
	Tier 4	11
	Tier 5	1
Chronotype groups (SIC)	Type 1	23
	Type 2	19
	Type 3	33
	Type 4	14
	Type 5	25

The distribution of the categorizing variables is also displayed. The values of the performance levels as defined by McKay et al. (1) were changed to increase the clarity of the table (every value was increased by 1) (Acc. = accuracy; BMI = body mass index; BSS = Brief Self Control Scale; IPAQ = International Physical Activity Questionnaire; MET = metabolic equivalent of task; MT = Mental Toughness Questionnaire; PVT = Psychomotor Vigilance Test; RT = reaction time; SART = Sustained Attention to Response Task; SIC = single item chronotype scale; TAI = Trait Anxiety Inventory; $\dot{V}O_{2max}$ = maximal oxygen uptake).

1. McKay AKA, Stellingwerff T, Smith ES, Martin DT, Mujika I, Goosey-Tolfrey VL, et al. Defining training and performance caliber: a participant classification framework. *Int J Sports Physiol Perf*. 2021:1–15. doi: 10.1123/ijspp.2021-0451.

comparing pre- to postvalues in the control condition ($t = 0.91$; $P = 0.367$; $d = 0.09$). However, there was a significant worsening of RT in the Stroop task condition from pre- to postvalues ($t = -5.40$; $P < 0.001$; $d = 0.55$), and a significant difference in postvalues between task conditions ($t = 6.55$; $P < 0.001$; $d = 0.65$), with a worse RT in the Stroop task condition compared with the documentary condition.

Both the statistical models of the ACC of the Go ($\beta = -0.33$; $P < 0.001$; 95% CI [−0.51, −0.15]; SE = 0.09) and NoGo ($\beta = -0.23$; $P = 0.017$; 95% CI [−0.42, −0.04]; SE = 0.10) stimuli showed a significant main effect of time, indicating an overall decrease in ACC over time, regardless of task exposure. No significant main effects of task conditions or interaction effects between task and time were found for ACC outcomes related to cognitive performance fatigability.

A significant interaction effect was found between task and time for the motivation ($P = 0.013$) and mental workload ($P < 0.001$) linked to the cognitive performance task. At baseline, there was no significant difference between tasks in motivation ($t = 0.41$; $P = 0.684$; $d = 0.04$). However, the motivation to perform the post-GoNoGo task was significantly higher in the control condition compared with the intervention condition ($t = -3.38$; $P = 0.001$; $d = 0.34$). Moreover, the motivation decreased from pre- to post-GoNoGo task in both the intervention ($t = -8.45$; $P < 0.001$; $d = 0.85$) and the control ($W = 3883.5$; $P < 0.001$; $r = 0.41$) condition. *Post hoc* analyses of the NASA-TLX revealed no significant baseline difference between task conditions ($t = 0.30$; $P = 0.765$; $d = 0.03$), nor a significant difference when comparing pre- to post-GoNoGo tasks in the control condition ($t = -0.98$; $P = 0.3318$; $d = 0.10$). However, there was a significant increase from pre- to post-task in the intervention condition ($t = -5.90$; $P < 0.001$; $d = 0.60$), and a significantly higher workload value for the post-GoNoGo task in the intervention condition compared with the control condition ($t = 5.83$; $P < 0.001$; $d = 0.58$).

Effects on physical performance fatigability. An overview of the end results of the time trial in both task conditions is depicted in Figure 4. Figure 5 displays the pacing strategy of the participants, and Figure 6 shows an overview of the changes in subjective values during the time trial.

There was a significant negative influence of Stroop task exposure on time trial cadence ($t = -2.06$; $P = 0.042$; 95% CI [−2.72, −0.05]; $d = 0.22$), and a trend toward a significant negative effect of Stroop task exposure on time trial distance ($t = -1.92$; $P = 0.059$; 95% CI [−0.38, 0.01]; $d = 0.20$). Analyses showed no evidence of a negative influence of the intervention task with regard to the workload, maximal power, and mean power.

Analyses of pacing values showed a significant increase in power output ($F = 81.14$; $P < 0.001$; $\eta_p^2 = 0.29$) and cadence ($F = 73.87$; $P < 0.001$; $\eta_p^2 = 0.27$) over time. In addition, exposure to the Stroop task was associated with a significant decrease in cadence during the time trial ($F = 14.25$; $P < 0.001$; 95% CI [−2.52, −0.80]; $\eta_p^2 = 0.01$). No main effect of task was found for power output, and no significant interaction effects were observed.

There was a significant effect of task and time on the feeling of physical fatigue (task: $F = 14.28$; $P < 0.001$; 95% CI [−0.34, −0.12]; $\eta_p^2 = 0.02$; time: $F = 459.67$; $P < 0.001$; $\eta_p^2 = 0.69$) and RPE (task: $F = 9.90$; $P = 0.002$; 95% CI

TABLE 2. Overview of descriptive values and statistical models of all primary and related outcomes.

	Descriptive Values		Statistics			
Outcome	**Intervention**	**Control**	**Effect**	**TS**	***P***	**ES**
Physical performance						
• Distance (km)	12.6 ± 1.8	12.8 ± 1.7	*Task*	–1.92 (*t*)	*0.059*	0.20 (*d*)
• Mean cadence (rpm)	87 ± 12	88 ± 12	*Task*	–2.06 (*t*)	**0.042**	0.22 (*d*)
• Mean workload (kJ)	163.6 ± 51.3	164.2 ± 48.2	*Task*	–0.33 (*t*)	0.741	0.03 (*d*)
• Mean power (W)	180 ± 57	183 ± 54	*Task*	–1.15 (*t*)	0.252	0.12 (*d*)
• Max power (W)	265 ± 129	249 ± 97	*Task*	1.25 (*t*)	0.216	0.13 (*d*)
• Pacing power (W; 3 min av.)	1: 173 ± 54	1: 171 ± 48	*Task*	0.68 (*F*)	0.408	0.001 (η_p^2)
	2: 175 ± 55	2: 175 ± 52	*Time*	81.14 (*F*)	**<0.001**	0.29 (η_p^2)
	3: 177 ± 56	3: 178 ± 54	*Int.*	1.46 (*F*)	0.201	0.01 (η_p^2)
	4: 183 ± 61	4: 186 ± 57				
	5: 200 ± 68	5: 202 ± 66				
	6: 251 ± 133	6: 235 ± 100				
• Pacing cadence (rpm; 3 min av.)	1: 82 ± 13	1: 83 ± 12	*Task*	14.25 (*F*)	**<0.001**	0.01 (η_p^2)
	2: 85 ± 13	2: 86 ± 12	*Time*	73.87 (*F*)	**<0.001**	0.27 (η_p^2)
	3: 88 ± 13	3: 89 ± 12	*Int.*	0.45 (*F*)	0.815	0.002 (η_p^2)
	4: 90 ± 13	4: 92 ± 13				
	5: 92 ± 13	5: 94 ± 13				
	6: 93 ± 17	6: 96 ± 17				
• Physical fatigue feeling (AU)	WU: 20 ± 15	WU: 17 ± 14	*Task*	14.28 (*F*)	**<0.001**	0.02 (η_p^2)
	3 min: 36 ± 15	3 min: 32 ± 14	*Time*	459.67 (*F*)	**<0.001**	0.69 (η_p^2)
	6 min: 48 ± 15	6 min: 43 ± 14	*Int.*	0.59 (*F*)	0.672	0.003 (η_p^2)
	9 min: 58 ± 15	9 min: 55 ± 14				
	12 min: 67 ± 16	12 min: 65 ± 16				
	15 min: 76 ± 17	15 min: 76 ± 16				
	CD: 61 ± 21	CD: 59 ± 21				
• Mental fatigue feeling (AU)	WU: 37 ± 22	WU: 20 ± 15	*Task*			
	3 min: 36 ± 20	3 min: 23 ± 16	•WU	2615 (*W*)	**<0.001**	0.47 (*r*)
	6 min: 37 ± 20	6 min: 25 ± 17	•3 min	2356 (*W*)	**<0.001**	0.46 (*r*)
	9 min: 36 ± 21	9 min: 26 ± 18	•6 min	2423 (*W*)	**<0.001**	0.42 (*r*)
	12 min: 36 ± 23	12 min: 29 ± 19	•9 min	2537.5 (*W*)	**<0.001**	0.39 (*r*)
	15 min: 39 ± 24	15 min: 32 ± 22	•12 min	2324 (*W*)	**<0.001**	0.33 (*r*)
	CD: 32 ± 23	CD: 25 ± 18	•15 min	2448.5 (*W*)	**<0.001**	0.34 (*r*)
			Time			
			•Stroop	6.78 (χ^2)	0.238	0.02 (*W*)
			•Docu	117.83 (χ^2)	**<0.001**	0.26 (*W*)
• RPE (AU)	WU: 15 ± 11	WU: 14 ± 11	*Task*	9.90 (*F*)	**0.002**	0.01 (η_p^2)
	3 min: 42 ± 16	3 min: 39 ± 15	*Time*	719.30 (*F*)	**<0.001**	0.78 (η_p^2)
	6 min: 52 ± 15	6 min: 50 ± 15	*Int.*	0.15 (*F*)	0.981	0.001 (η_p^2)
	9 min: 61 ± 16	9 min: 58 ± 15				
	12 min: 70 ± 16	12 min: 69 ± 15				
	15 min: 80 ± 17	15 min: 78 ± 16				
	CD: 28 ± 17	CD: 28 ± 18				
• Moti-VAS (AU)	69 ± 17	71 ± 18	*Task*	1.58 (*t*)	0.117	0.17 (*d*)
• HR (bpm)	Baseline: 68 ± 13	Baseline: 69 ± 12	*Task*	0.08 (*F*)	0.785	<0.001 (η_p^2)
	WU: 104 ± 16	WU: 103 ± 17	*Time*	2890.83 (*F*)	**<0.001**	0.94 (η_p^2)
	3 min: 143 ± 17	3 min: 143 ± 17	*Int.*	0.08 (*F*)	0.998	<0.001 (η_p^2)
	6 min: 152 ± 15	6 min: 152 ± 16				
	9 min: 159 ± 15	9 min: 159 ± 15				
	12 min: 164 ± 14	12 min: 165 ± 13				
	15 min: 170 ± 14	15 min: 171 ± 12				
	CD: 139 ± 17	CD: 139 ± 2				
• Blood lactate (mmol·L^{-1})	PRE: 1.85 ± 0.53	PRE: 1.78 ± 0.50	*Task*	1.12 (*F*)	0.291	0.004 (η_p^2)
	POST: 8.49 ± 2.91	POST: 8.98 ± 2.86	*Time*	1182.10 (*F*)	**<0.001**	0.82 (η_p^2)
			Int.	1.90 (*F*)	0.170	0.007 (η_p^2)
• Blood glucose (mg·dL^{-1})	PRE: 108.61 ± 11.50	PRE: 107.42 ± 12.57	*Task*	0.12 (*F*)	0.735	<0.001 (η_p^2)
	POST: 105.28 ± 16.17	POST: 106.42 ± 21.13	*Time*	4.58 (*F*)	**0.034**	0.02 (η_p^2)
			Int.	0.32 (*F*)	0.571	0.001 (ηp2)
Cognitive performance						
• Go RT (ms)	PRE: 367.79 ± 29.46	PRE: 368.73 ± 31.14	*Task*	17.60 (*F*)	**<0.001**	0.06 (η_p^2)
	POST: 380.45 ± 33.59	POST: 367.30 ± 31.39	*Time*	14.89 (*F*)	**<0.001**	0.05 (η_p^2)
			Int.	23.44 (*F*)	**<0.001**	0.07 (η_p^2)
• Go ACC (ratio; median ± IQR)	PRE: 0.99 ± 0.01	PRE: 0.99 ± 0.01	*Task*	0.07 (β)	0.520	0.10 (SE)
	POST: 0.99 ± 0.02	POST: 0.99 ± 0.01	*Time*	–0.33 (β)	**<0.001**	0.09 (SE)
			Int.	0.17 (β)	0.213	0.13 (SE)

Continued next page

TABLE 2. *(Continued)*

Outcome	Descriptive Values: Intervention	Descriptive Values: Control	Statistics: Effect	TS	*P*	ES
• NoGo ACC (ratio: median ± IQR)	PRE: 0.94 ± 0.11	PRE: 0.92 ± 0.08	*Task*	0.11 (β)	0.295	0.11 (SE)
	POST: 0.92 ± 0.11	POST: 0.92 ± 0.08	*Time*	−0.23 (β)	**0.017**	0.10 (SE)
			Int.	0.14 (β)	0.343	0.14 (SE)
• Moti-VAS (AU)	PRE: 75 ± 14	PRE: 75 ± 16	*Task*	7.54 (*F*)	**0.007**	0.02 (η_p^2)
	POST: 61 ± 20	POST: 67 ± 19	*Time*	102.87 (*F*)	**<0.001**	0.26 (η_p^2)
			Int.	4.87 (*F*)	**0.028**	0.02 (η_p^2)
• NASA-TLX (sum score AU)	PRE: 43 ± 18	PRE: 42 ± 16	*Task*	19.53 (*F*)	**<0.001**	0.06 (η_p^2)
	POST: 53 ± 19	POST: 44 ± 18	*Time*	25.64 (*F*)	**<0.001**	0.08 (η_p^2)
			Int.	15.95 (*F*)	**<0.001**	0.05 (η_p^2)
Additional secondary outcomes						
• BRUMS Vigor (AU)	PRE: 9 ± 3	PRE: 9 ± 3	*Task*			
	POST: 6 ± 3	POST: 7 ± 4	•PRE	1368 (*W*)	0.622	0.04 (*r*)
			•POST	576 (*W*)	**<0.001**	0.34 (*r*)
			Time			
			•Stroop	3544 (*W*)	**<0.001**	0.56 (*r*)
			•Docu	2482.5 (*W*)	**<0.001**	0.42 (*r*)
• BRUMS Fatigue (AU)	PRE: 3 ± 3	PRE: 2 ± 2	*Task*			
	POST: 6 ± 4	POST: 3 ± 2	•PRE	1812.5 (*W*)	**0.020**	0.17 (*r*)
			•POST	3948 (*W*)	**<0.001**	0.55 (*r*)
			Time			
			•Stroop	109 (*W*)	**<0.001**	0.55 (*r*)
			•Docu	670.5 (*W*)	**0.001**	0.23 (*r*)
• BRUMS Tension (AU, median ± IQR)	PRE: 0 ± 1	PRE: 0 ± 1	*Task*	−0.002 (β)	0.988	0.14 (SE)
	POST: 0 ± 1	POST: 0 ± 0	*Time*	−0.19 (β)	0.187	0.15 (SE)
			Int.	−0.52 (β)	**0.038**	0.25 (SE)
• BRUMS Confusion (AU, median ± IQR)	PRE: 0 ± 0	PRE: 0 ± 1	*Task*	0.10 (β)	0.571	0.18 (SE)
	POST: 0 ± 2	POST: 0 ± 0	*Time*	0.34 (β)	**0.047**	0.17 (SE)
			Int.	−1.10 (β)	**<0.001**	0.32 (SE)
• BRUMS Anger (AU, median ± IQR)	PRE: 0 ± 1	PRE: 0 ± 1	*Task*	−0.11 (β)	0.494	0.16 (SE)
	POST: 1 ± 2	POST: 0 ± 1	*Time*	0.80 (β)	**<0.001**	0.13 (SE)
			Int.	−0.72 (β)	**<0.001**	0.20 (SE)
• BRUMS Depression (AU, median ± IQR)	PRE: 0 ± 0	PRE: 0 ± 0	*Task*	−0.08 (β)	0.727	0.22 (SE)
	POST: 0 ± 0	POST: 0 ± 0	*Time*	0.26 (β)	0.390	0.30 (SE)
			Int.	−0.38 (β)	0.388	0.44 (SE)
• KSS (AU)	PRE: 4 ± 2	PRE: 4 ± 2	*Task*	−0.51 (β)	**0.046**	0.26 (SE)
	POST: 6 ± 2	POST: 5 ± 2	*Time*	1.54 (β)	**<0.001**	0.26 (SE)
			Int.	−0.62 (β)	*0.086*	0.36 (SE)
• B-VAS (AU)	67 ± 22	25 ± 20	*Task*	4604.5 (*W*)	**<0.001**	0.58 (*r*)
Manipulation checks						
• MVAS base (AU)	18 ± 15	19 ± 17	*Task*	2334.5 (*W*)	0.569	0.04 (*r*)
• MVAS int/con (AU)	PRE: 23 ± 16	PRE: 24 ± 17	*Task*			
	POST: 69 ± 19	POST: 28 ± 20	•PRE	2329.5 (*W*)	0.716	0.03 (*r*)
			•POST	4719 (*W*)	**<0.001**	0.60 (*r*)
			Time			
			•Stroop	11 (*W*)	**<0.001**	0.61 (*r*)
			•Docu	1794 (*W*)	*0.099*	0.12 (*r*)
• MVAS preTT (AU)	59 ± 19	35 ± 19	*Task*	4567.5 (*W*)	**<0.001**	0.54 (*r*)
	Normal stimuli	*Inhibitory*				
• Stroop RT (ms)	Block 1: 590.48 ± 97.03	Block 1: 526.74 ± 151.68	*Type*	40.65 (*F*)	**<0.001**	0.08 (η_p^2)
	Block 2: 602.90 ± 97.75	Block 2: 576.22 ± 138.63	*Block*	17.58 (*F*)	**<0.001**	0.07 (η_p^2)
	Block 3: 598.92 ± 94.76	Block 3: 588.40 ± 125.89	*Int.*	8.91 (*F*)	**<0.001**	0.04 (η_p^2)
• Stroop ACC (ratio)	Block 1: 0.83 ± 0.10	Block 1: 0.60 ± 0.18	*Type*			
	Block 2: 0.86 ± 0.09	Block 2: 0.70 ± 0.18	•1	4654 (*W*)	**<0.001**	0.61 (*r*)
	Block 3: 0.86 ± 0.10	Block 3: 0.73 ± 0.17	•2	4174 (*W*)	**<0.001**	0.57 (*r*)
			•3	4171 (*W*)	**<0.001**	0.57 (*r*)
			Block			
			•Norm.	27.77 (χ^2)	**<0.001**	0.14 (*W*)
			•Inh.	74.58 (χ^2)	**<0.001**	0.39 (*W*)

Values in bold represent a significant effect. All descriptive values are mean ± SD unless stated otherwise (β = estimate; η_p^2 = partial eta square; ACC = accuracy; av. = average; AU = arbitrary units; CD = cool-down; CON = control; *d* = Cohen's *d*; Inh. = inhibitory; IQR = interquartile range; Norm. = normal; *r* = Cohen's *r*; RT = reaction time; sqrt = square root transformation; SE = standard error; TS = test statistic; *W* = *W* value Wilcoxon test; WU = warm-up).

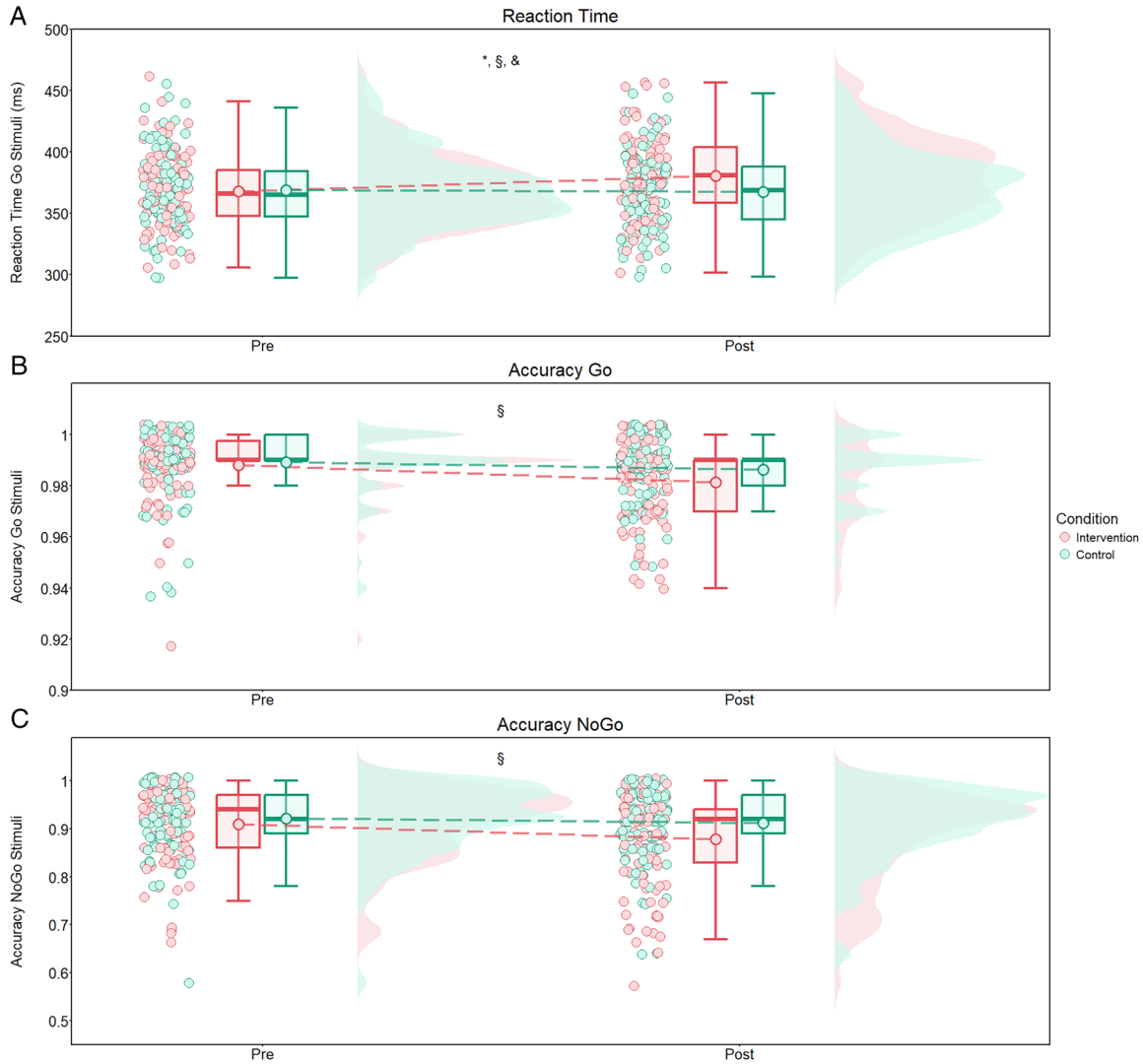

FIGURE 3—Overview of the difference between conditions for all performance outcomes collected during the GoNoGo task using individual data points, boxplots, and raincloud plots. (A) Reaction time on the Go stimuli; (B) accuracy on the Go stimuli; (C) accuracy on the NoGo stimuli. Points in the middle of the boxplot represent means, whereas the middle of the boxplot represents the median. *Red* represents the intervention condition, whereas *green* represents the control condition. * = significant main effect of condition; § = significant main effect of time; & = significant interaction effect between condition and time.

[−0.22, −0.05]; $\eta_p^2 = 0.01$; time: $F = 719.30$; $P < 0.001$; $\eta_p^2 = 0.78$) taken during the time trial, with higher values related to exposure to the intervention task, and a clear increase in values over time. No interaction effects were observed in both scales. Nonparametric Friedman analyses of the feeling of mental fatigue during the time trial showed a significant increase over time in the control session ($\chi^2 = 117.83$; $P < 0.001$; $W = 0.26$), but no significant effect related to the intervention task. Wilcoxon analyses of the same scale showed that the subjective feeling of mental fatigue remained higher in the Stroop task condition compared with the documentary condition throughout the time trial (all $P < 0.001$).

There was no influence of Stroop task exposure on motivation, blood lactate, blood glucose, and HR measures taken during the time trial. The HR and blood lactate values significantly increased over time (HR: $F = 2890.83$; $P < 0.001$; $\eta_p^2 = 0.94$; lactate: $F = 1182.10$; $P < 0.001$; 95% CI [−7.31, −6.52]; $\eta_p^2 = 0.82$), whereas blood glucose significantly decreased over time ($F = 4.58$; $P = 0.034$; 95% CI [0.001, 0.02]; $\eta_p^2 = 0.02$).

Outcomes Linked to the Intervention/Control Task

There was no significant difference between task conditions for baseline mental fatigue visual analogue scale values before the start of the intervention/control task, and no significant effect of time within the control task condition. Analysis showed a significant increase in the subjective level of mental fatigue before compared with after the Stroop task ($W = 11$; $P < 0.001$; 95% CI [−51.50, −42.50]; $r = 0.61$), which also resulted in a significantly higher mental fatigue visual analogue scale value in

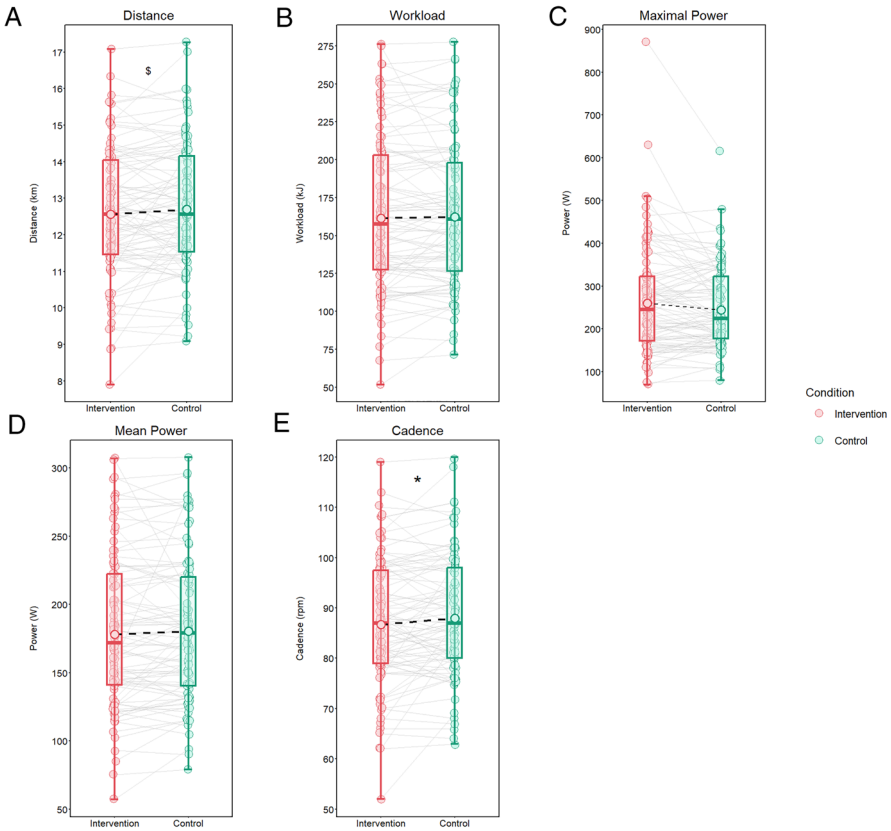

FIGURE 4—Overview of the difference between conditions for all performance outcomes collected during the time trial represented by boxplots and individual data points. (A) Distance; (B) workload; (C) maximal power; (D) mean power; (E) cadence, (F) speed. Points in the middle of the boxplot represent means, whereas the middle of the boxplot represents the median. *Red* represents the intervention condition, whereas *green* represents the control condition. * = significant main effect of condition; $ = trend toward significant main effect of condition.

the intervention compared with the control condition post-intervention/control task ($W=4719$; $P<0.001$; 95% CI [37.50, 47.50]; $r=0.60$). Further tests revealed that this significant difference between task groups persisted before the physical performance task ($W=4567.5$; $P<0.001$; 95% CI [19.50, 28.50]; $r=0.54$), meaning that participants felt a higher level of subjective mental fatigue before beginning the time trial.

Statistical analysis of Stroop RT revealed a significant interaction effect between block and stimulus type ($P<0.001$). Linear models conducted on both types of stimuli showed a significant effect of time in both types (normal: $F=4.27$; $P=0.015$; $\eta_p^2=0.04$/inhibitory: $F=25.76$; $P<0.001$; $\eta_p^2=0.21$). Detailed analyses of this effect of time revealed a significant increase in RT between block 1 and block 2 regardless of type (normal: $t=-2.799$; $P=0.006$; $d=0.29$/inhibitory: $t=-5.36$; $P<0.001$; $d=0.55$). However, the increase in RT between blocks 1 and 3 was only significant in the inhibitory stimuli ($t=-6.01$; $P<0.001$; $d=0.61$). There were no significant differences between block 2 and block 3. *Post hoc* paired samples t tests showed a significantly worse reaction time when comparing types at block 1 ($t=-5.86$; $P<0.001$; $d=0.60$) and block 2 ($t=-2.78$; $P=0.007$; $d=0.28$). Analyses of the delta calculation showed a significant decrease in the difference between the normal and inhibitory stimuli as time went on ($F=25.72$; $P<0.001$; $\eta_p^2=0.21$).

Friedman tests performed on the ACC of the Stroop task revealed a significant improvement over time in both stimuli types (all $P<0.001$). Wilcoxon tests demonstrated that this improvement was present across all blocks in the inhibitory stimuli (1 vs. 2: $W=454.5$; $P<0.001$; $r=0.48$; 2 vs. 3: $W=1197.5$; $P<0.001$; $r=0.24$; 1 vs. 3: $W=364$; $P<0.001$; $r=0.51$). For the normal stimuli, there were only significant improvements between the first and the second ($V=1004$; $P<0.001$; $r=0.26$), and the first and third ($V=944$; $P<0.001$; $r=0.30$) block of the Stroop task. Meanwhile, the ACC of the normal stimuli was significantly higher across all blocks compared with the inhibitory stimuli. The model examining the Stroop effect showed a decrease in the difference in ACC over time ($F=57.85$; $P<0.001$; $\eta_p^2=0.38$).

Analyses of the BRUMS fatigue subscale showed a significant increase over time in both task conditions (Stroop: $W=109$; $P<0.001$; 95% CI [−4.50, −3.50]; $r=0.55$; Docu: $W=670.5$; $P=0.001$; 95% CI [−1.50, −0.50]; $r=0.23$) and a significant increase in the value on

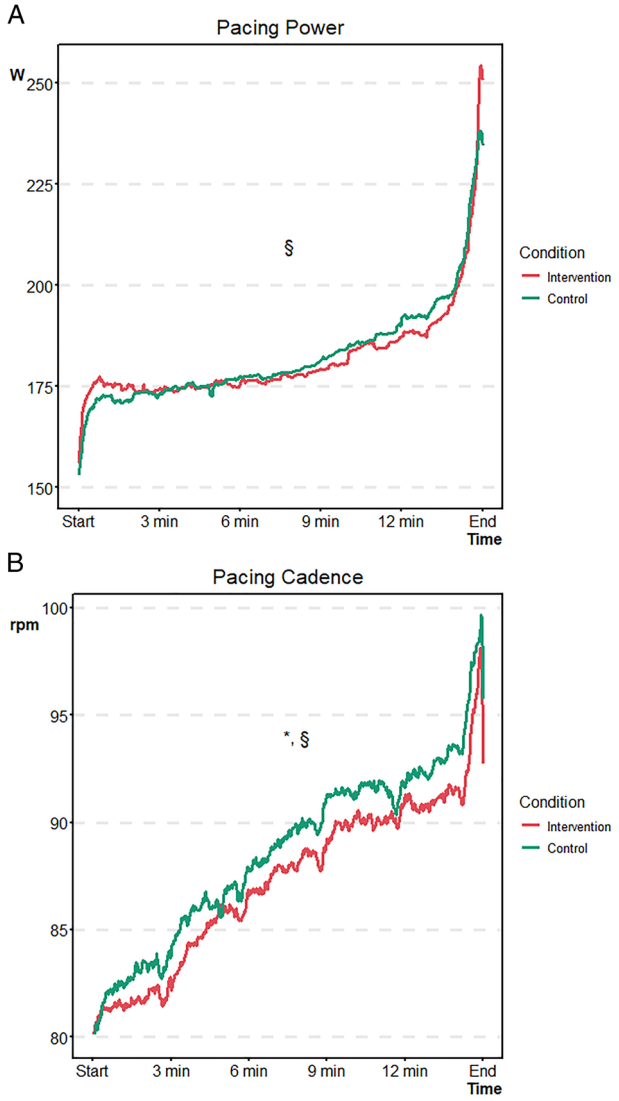

FIGURE 5—Overview of the pacing evolution over time during the physical performance task. (A) Power output; (B) cadence. *Red* represents the intervention condition, whereas *green* represents the control condition. The warm-up and cool-down are not included in the graphs to provide a clearer visualization of the data. The limits of the values on the *y* axis are altered for the same reason. * = significant main effect of condition; § = significant main effect of time..

the fatigue subscale in the intervention condition compared with the control condition in both measurement periods (pre: $W = 1812.5$; $P = 0.020$; 95% CI [0.00003, 1.00]; $r = 0.17$; post: $W = 3948$; $P < 0.001$; 95% CI [3.00, 4.00]; $r = 0.55$). Meanwhile, baseline values of the Vigor subscale portrayed no significant difference between conditions. However, there was a significantly higher level of vigor in the control condition compared with the intervention condition post-task ($W = 576$; $P < 0.001$; 95% CI [−3.00, −1.50]; $r = 0.34$). In both task conditions, the level of vigor decreased significantly over time (all $P < 0.001$). Different models showed a significant interaction effect between task and time for the tension ($P = 0.038$), anger ($P < 0.001$), and confusion ($P < 0.001$) subscales of the BRUMS questionnaire. These interactions were not further examined because of the very limited difference in median values. There was no influence of task, time, or an interaction effect between task and time for the depression subscale.

Analysis also showed a significantly higher level of boredom related to the Stroop task compared with the feeling induced by the documentary ($W = 4604.5$; $P < 0.001$; 95% CI [36.50, 49.00]; $r = 0.58$). Regarding sleepiness, there was a significant overall higher sleepiness value in the Stroop task condition compared with the documentary condition ($\beta = -0.51$; $P = 0.046$; 95% CI [−1.02, −0.01]; SE = 0.26), and an overall increase in

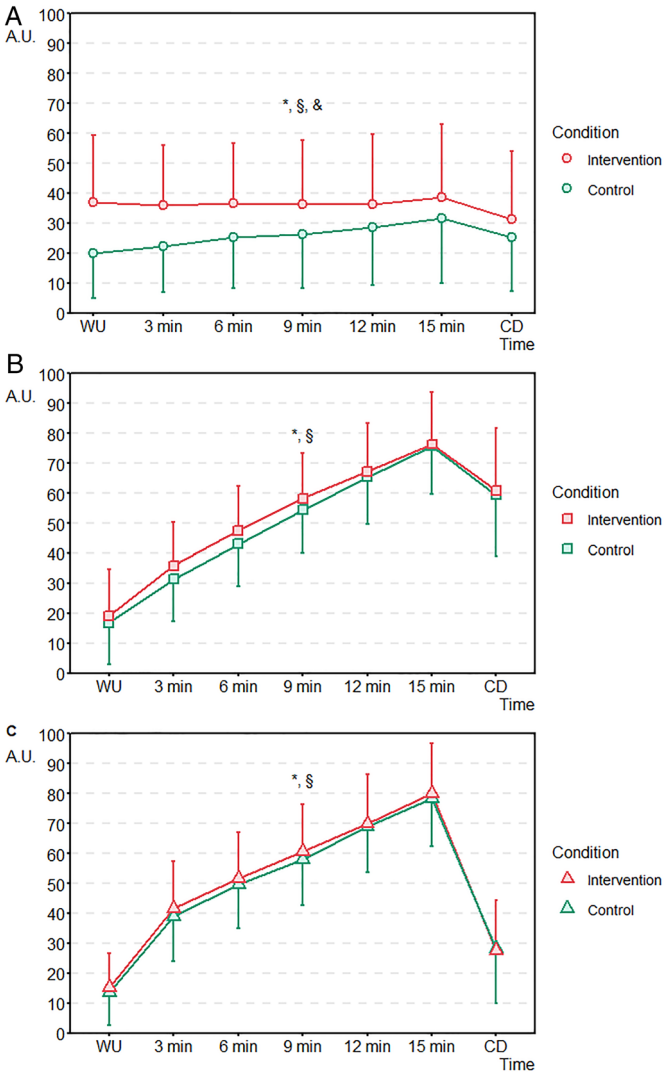

FIGURE 6—Overview of the difference between conditions for the verbal subjective feelings of mental fatigue, physical fatigue, and the RPE during the time trial. (A) Mental fatigue; (B) physical fatigue; (C) RPE. *Red* represents the intervention condition, whereas *green* represents the control condition. * = significant main effect of condition; § = significant main effect of time; & = significant interaction effect between condition and time (AU = arbitrary units; CD = cool-down; WU = warm-up).

sleepiness over time regardless of task ($\beta = 1.54$; $P < 0.001$; 95% CI [1.03, 2.07]; SE = 0.26). There was no interaction between task and time for the sleepiness values.

DISCUSSION

Summary of findings. The aim of the present study was to investigate the effects of prolonged intense cognitive exposure on cognitive and physical performance fatigability, which has been defined by some as the behavioral effects of mental fatigue. We also addressed methodological limitations by optimizing the way this type of fatigue is induced, selecting the most sensitive outcome measures and detection techniques, and ensuring transparent reporting. We hypothesized that the Stroop task would impair physical and cognitive performance fatigability. In line with our hypothesis, the intervention condition led to slower GoNoGo reaction times and reduced time trial cadence. GoNoGo task-related motivation and mental workload were also impaired after the Stroop task. The intervention condition further led to an increase in all subjective responses (i.e., feelings of fatigue and RPE) collected during the time trial. Notably, there was also an increase in the feeling of boredom (B-VAS), sleepiness (KSS), general fatigue (BRUMS), and a decrease in vigor (BRUMS). Watching a documentary similarly increased sleepiness and general fatigue, and reduced vigor. The results of the Stroop task over time showed dynamic adaptations occurring in performance throughout the 45 min for the most complex stimuli.

Influence of a prolonged cognitive load on physical performance fatigability. Overall, results support previous findings indicating that the negative effects of cognitive exposure on physical performance are primarily mediated by psychophysiological mechanisms, related to the phenomenon of mental fatigue (11). This is reflected in the absence of significant effects on peripheral physiological outcome measures, and the significant differences in subjective values. Originally, it was suggested that changes in physical performance outcomes due to prolonged cognitive activity were primarily driven by an increase in RPE and a decrease in motivation (15,56). However, there was no difference in motivation before the time trial in the present study, and the influence of cognitive load on RPE, although significant, remained small. Although it is important to note that these results could be partially explained by the choice of measurement method (i.e., the task-related motivation [11], RPE 100 scale [38], and time-based time trial [57]), it seems that subjective feelings of fatigue appear to play a key role in understanding how cognitive load affects physical performance. Changes in subjective feelings of fatigue are thought to originate from multiple brain areas (58), further confirming the multifaceted nature of mental fatigue. It is especially notable that a purely cognitive load also consistently increased the feeling of physical fatigue during the time trial, indicating an inherent link between two constructs that are induced in very different manners. An additional finding concerns pacing strategies: cadence was consistently higher in the control condition. Research suggests that pacing strategies are the result of knowledge of the end point of the trial (the metabolic demands of the task), which are then weighed against the participants' metabolic capacity (57,59). As such, pacing strategies are the result of the anticipation of possible failure of physiological systems (60). Taken together, our findings suggest that prior cognitive load may influence anticipatory regulation during physical tasks, leading individuals to adopt a more conservative pacing strategy (61). This, along with the reported subjective experience, offers new insights into the ways prior prolonged cognitive exposure negatively affects physical performance fatigability.

Influence of a prolonged cognitive load on cognitive performance fatigability. Although there was a negative influence of the Stroop task on the RT of the GoNoGo task (compared with prevalues and the control condition), there seemed to be no effect on the ACC of either Go or NoGo stimuli. As the effects of prior cognitive load on subsequent cognitive performance fatigability are widespread (62) and include numerous executive functions (e.g., working memory [63], response inhibition [14,64] and attention [65,66]), this finding appears somewhat underwhelming. Interestingly, the study by Guo et al. (14) also found no difference in false alarm rates in the GoNoGo task after a mentally demanding task. The authors argue that this might have been because participants became more careful, prioritizing the maintenance of accuracy at the cost of higher reaction times (14). Our findings surrounding the Stroop interference effect also indicate that the 45-min duration might not have been sufficient to see the full development of cognitive performance adaptation. Results show a cost in RT for both types of stimuli, whereas this cost is higher for the more complex stimuli, and does not stabilize in terms of speed/accuracy trade-off. The fact that we see an increasing cost in RT for the more complex stimuli of the Stroop task, coupled with an increase in ACC of the Stroop task, suggests similar trade-off adjustments during the Stroop task as the study by Guo et al. (14). A study by the same authors, which looked at the influence of music on cognitive load effects, also found similar results (67). In that case, the lack of ACC differences was attributed to a possible floor effect, as participants maintained high levels of ACC throughout the trial (67). Similar explanations may apply to the present study. A closer look at our variables indicates that most participants were not particularly challenged in this GoNoGo task, as reflected by the low RT and high ACC scores. This indicates that participants engaged more automatic cognitive processes to achieve

task success, which are less impacted by mental fatigue (68). As task performance was still impacted by cognitive load, increasing task difficulty in the future and/or task duration of the induction could result in more clear-cut results regarding the cognitive performance decrement (44). These mixed results further emphasize the methodological difference in assessing physical and cognitive performance, and the need for a careful interpretation of cognitive variables, beyond a mere analysis of reaction times.

Presence of interindividual variability. A consistent finding across the results is that the influence of Stroop task exposure on the behavioral outcome measures, although significant, remains limited. For example, the effect of cognitive load on the distance of the time trial ($d = 0.20$) is smaller compared with meta-analyses examining cognitive exertion effects on endurance performance ($g = 0.35/0.32$) (10,19). Moreover, prior cognitive load did not affect other important physical performance parameters, such as workload and power output. The main reason for these results can arguably be found in Figure 4, which displays the difference between conditions for every outcome of the time trial. The graphs of time trial outcomes all show considerable variability in the way subjects dealt with the possible induction of mental fatigue. This variability is actually present throughout the trial, from the changes in cognitive performance outcomes to the increase in the subjective feeling of mental fatigue. A recent meta-analysis already emphasized that this variability is present across other studies investigating the impact of mental fatigue on physical endurance performance, which makes it difficult to clearly assess its effects and mechanisms (10). A meta-regression was conducted to see if different individual features (i.e., age, sex, training level, or BMI) impact this variability, but no significant influence was found (10). However, a meta-analysis is limited in its conclusions because of the large amount of heterogeneity in outcomes, population, and analysis methods across studies. Therefore, a final suggestion of this review was to conduct a large experimental trial that further investigates this variability and the link with individual features in one population (10). Although the current trial focuses on the overall behavioral and subjective effects of a prolonged cognitive load, a secondary analysis where the influence of the collected individual features is taken into account will provide invaluable information on the differing effects of prolonged cognitive load in different groups.

Future research. The present study offers several important directions for future research. First, it is important that future mental fatigue studies provide a nuanced view of their findings, and position their results within broader fatigue literature when possible. Second, the results indicate that further investigation into the effects of prior cognitive exposure on pacing strategies is warranted. This could further improve our understanding of neural mechanisms that underlie the negative effects of mental fatigue on physical performance fatigability (69). Moreover, the collected subjective results underscore the importance of incorporating measures of mental and physical fatigue in future studies. These measures could also provide practical insights for professionals working in performance-based fields. Third, evaluating cognitive performance encompasses more than applying random tasks before and after induction: the analysis of task parameters during induction offers valuable insights regarding the mechanistic investigation of mental fatigue. Finally, effort should be made to further investigate interindividual differences in the influence of cognitive exertion on human performance, along with the possible underlying individual features. A better understanding of these differences could significantly advance the field and contribute to a more nuanced model of fatigue.

CONCLUSIONS

The present study demonstrated that the completion of a 45-min Stroop task negatively influenced physical and cognitive performance fatigability in a sample of over 100 participants, possibly related to the concept of mental fatigue. Specifically, this prolonged cognitive exposure had a significant negative influence on the cadence, pacing strategy, and subjective responses (i.e., feelings of fatigue and RPE) during a 20-min cycling time trial. Moreover, reaction times on the GoNoGo task were significantly impaired. Secondary outcomes further showcase the complex nature of possible mental fatigue induction.

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Conflict of Interest: Jelle Habay, Y. Laurisa Arenales Arauz, Matthias Proost, Emilie Schampheleer, Elke Lathouwers, Kevin De Pauw, Nathalie Pattyn, Jeroen Van Cutsem, and Bart Roelands have no conflicts of interest of any type (i.e., financial, professional, or personal) relevant to the content of this study.

Compliance with Ethical Standards: The present study protocol and its procedures were approved by the Medical Ethics Committee of the UZ Brussel and the Vrije Universiteit Brussel, Belgium (B.U.N. 1432022000084), in accordance with the Declaration of Helsinki. All participants provided written informed consent to participate in the study. Jelle Habay, Y. Laurisa Arenales Arauz, Matthias Proost, Emilie Schampheleer, Elke Lathouwers, Kevin De Pauw, Nathalie Pattyn, Jeroen Van Cutsem, and Bart Roelands declare that the investigation and its reporting comply with all ethical standards.

Data Availability Statement: All data generated or analyzed during this study are included in this published article and its supplementary information files.

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